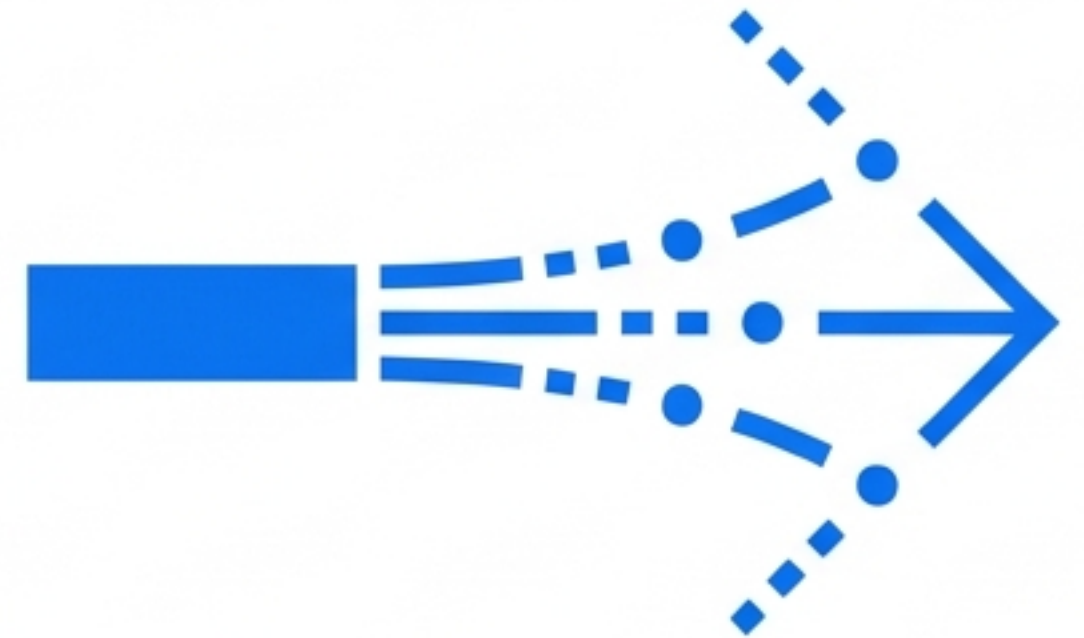


Evolutionary Trajectory Fine-Tuning

Reverse-Engineering Algorithmic Progression

Towards an Open-Loop Autonomous R&D Ecosystem and
General Innovation Accelerator.

Lukas Benda | April 2026



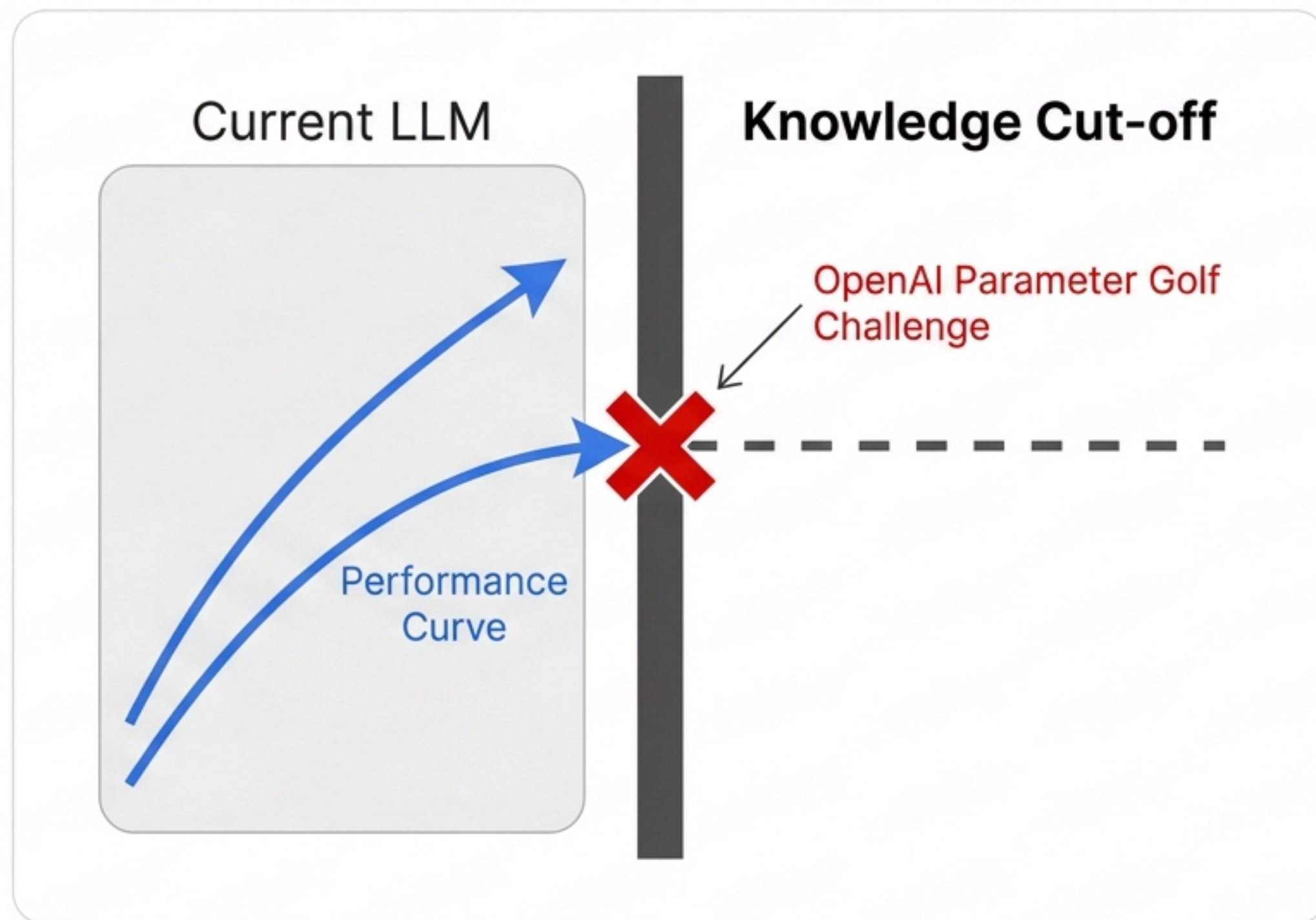
The Parametric Blindspot

Core Concept

Current LLMs operate as static code generators trapped by their pre-trained weights. They understand how code functions, but not how or why it evolves.

The Limitation

Under extreme resource constraints, models cannot autonomously synthesize a true SOTA+1 hypothesis relying solely on outdated parametric memory.

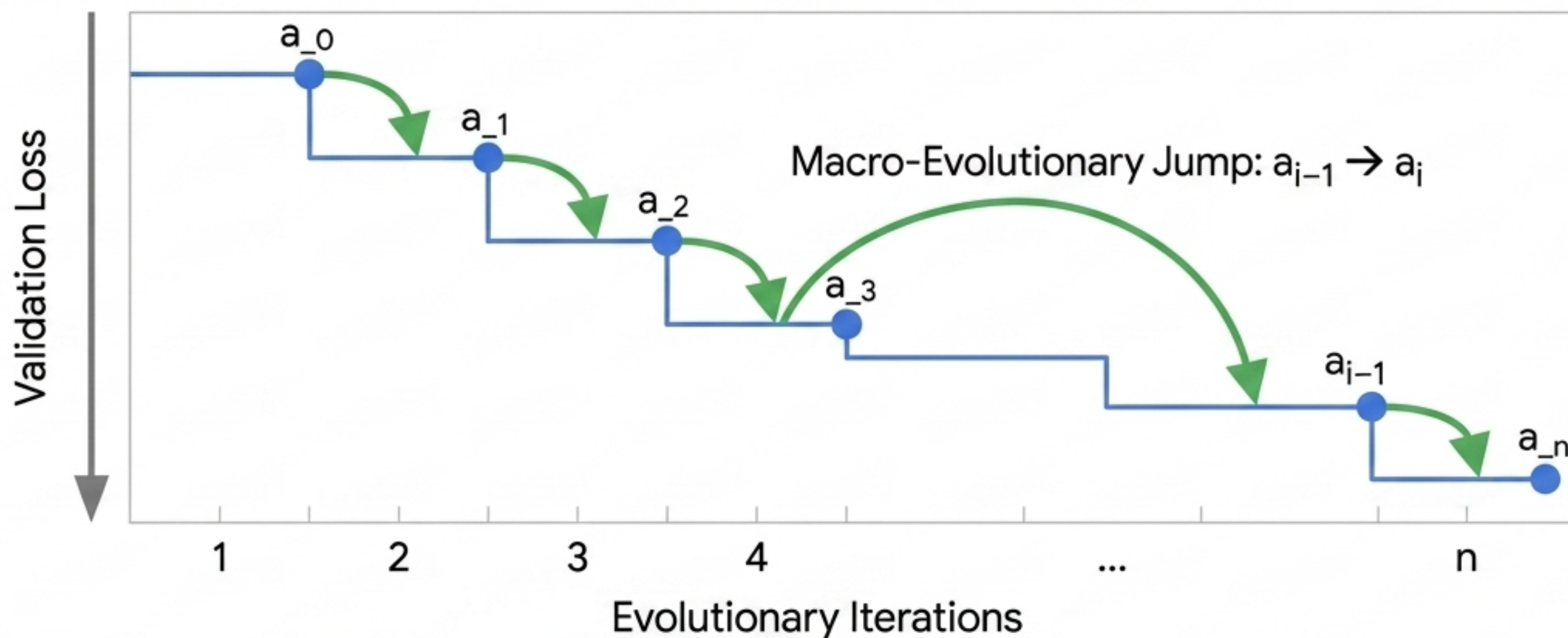


The Generative Paradigm Shift

	Current LLMs	ETFT Ecosystem
Core Role	Static Code Generator	Autonomous Research Scientist
Knowledge Bound	Frozen Pre-trained Weights	Open-Loop Agentic Framework
Training Focus	Raw Functional Execution	Evolutionary Trajectory Optimization
Failure Handling	Dead-ends & Hallucinations	Organic In-Context Reinforcement Learning

The Discrete Gradient

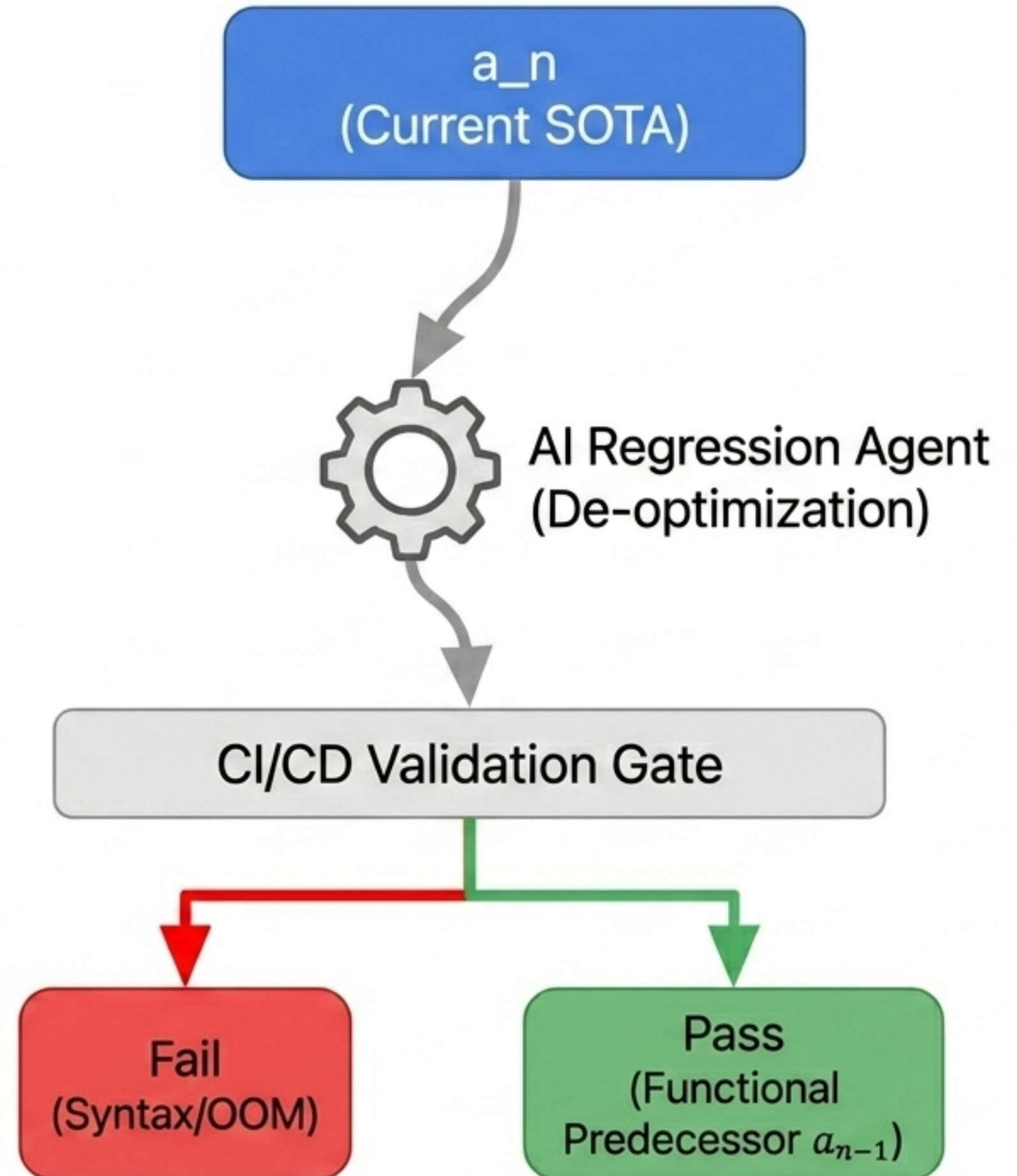
Let T = ordered sequence $(a_0, a_1, \dots, a_n)$ where objective fitness $F(a_{i-1}) < F(a_i)$.



Takeaway: ETFT does not train on disconnected code fragments. It trains on functional regressions—teaching the model the fundamental vectors of architectural innovation.

AI-Assisted Top-Down Regression Pipeline

Core Concept: Bottom-up manual curation is a scalability bottleneck. ETFT automates training data by de-optimizing SOTA algorithms subject to strict CI/CD validation.



Deconstructing the Evolutionary Corpus

D_Gen

(Generative Pairs)

Function: Maps the lower-complexity algorithm to its advanced successor.

Input: $a_{i-1} \rightarrow$ **Output:** a_i

D_Rationale

(Structural Meta-data)

Function: Translates code deltas into causal reasoning. Provides a comprehensive textual breakdown of the architectural shift.

D_Perf

(Performance Metrics)

Function: Hardware-constrained performance mapping to inform future in-silico triage.

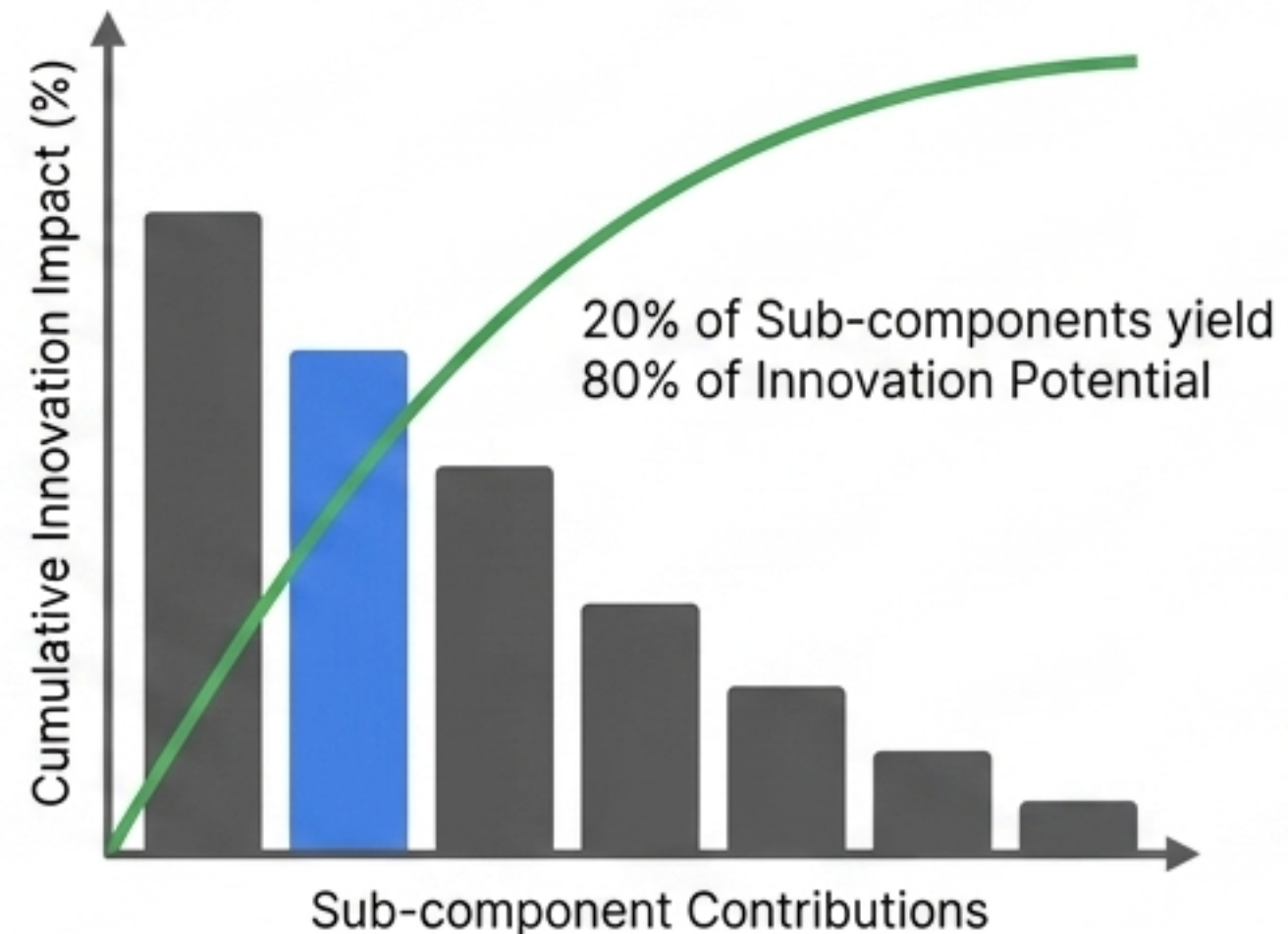
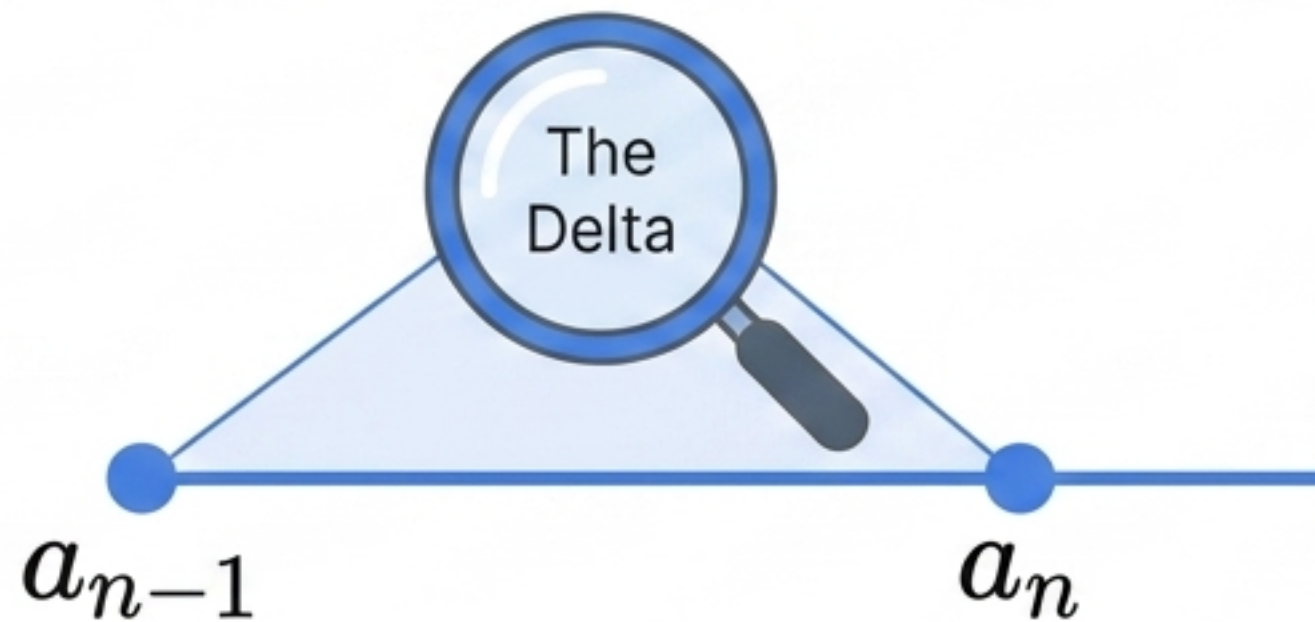
Retrospective Pareto (80/20) Delta Analysis

The Pivot

LLMs struggle with highly error-prone, real-time computational profiling.

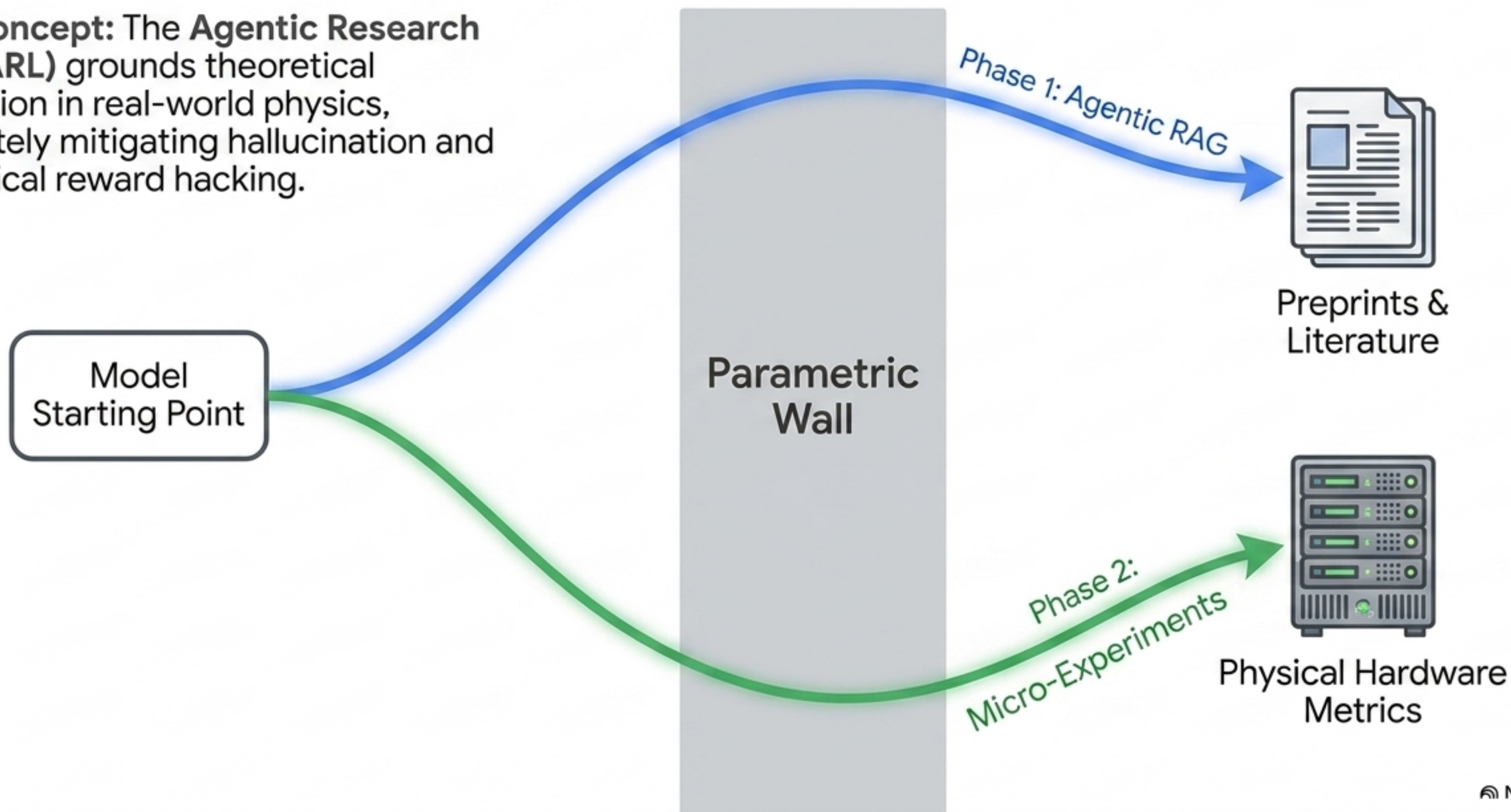
The Solution

Deterministic analysis of historical deltas. By quantifying the character and scope of past innovations, the system learns the patterns of success to project focus onto the future $a_{n-1} \rightarrow a_{n+1} \xrightarrow{\frac{1}{2}} a_{n+1}$ trajectory.



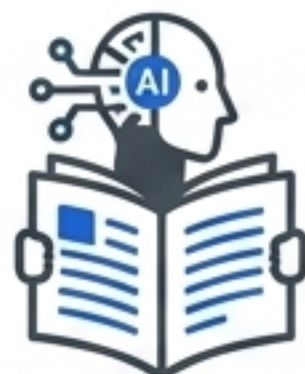
Overcoming the Parametric Wall

Core Concept: The **Agentic Research Loop (ARL)** grounds theoretical generation in real-world physics, completely mitigating hallucination and theoretical reward hacking.



The Agentic Research Loop (ARL)

Process Detail: Once the 80/20 bottlenecks are identified, the model formulates targeted research questions, triggering a two-phase autonomous deployment.



Node 1

Phase 1: Deep Literature Synthesis

Scours up-to-date scientific knowledge regarding identified bottlenecks.



Node 2

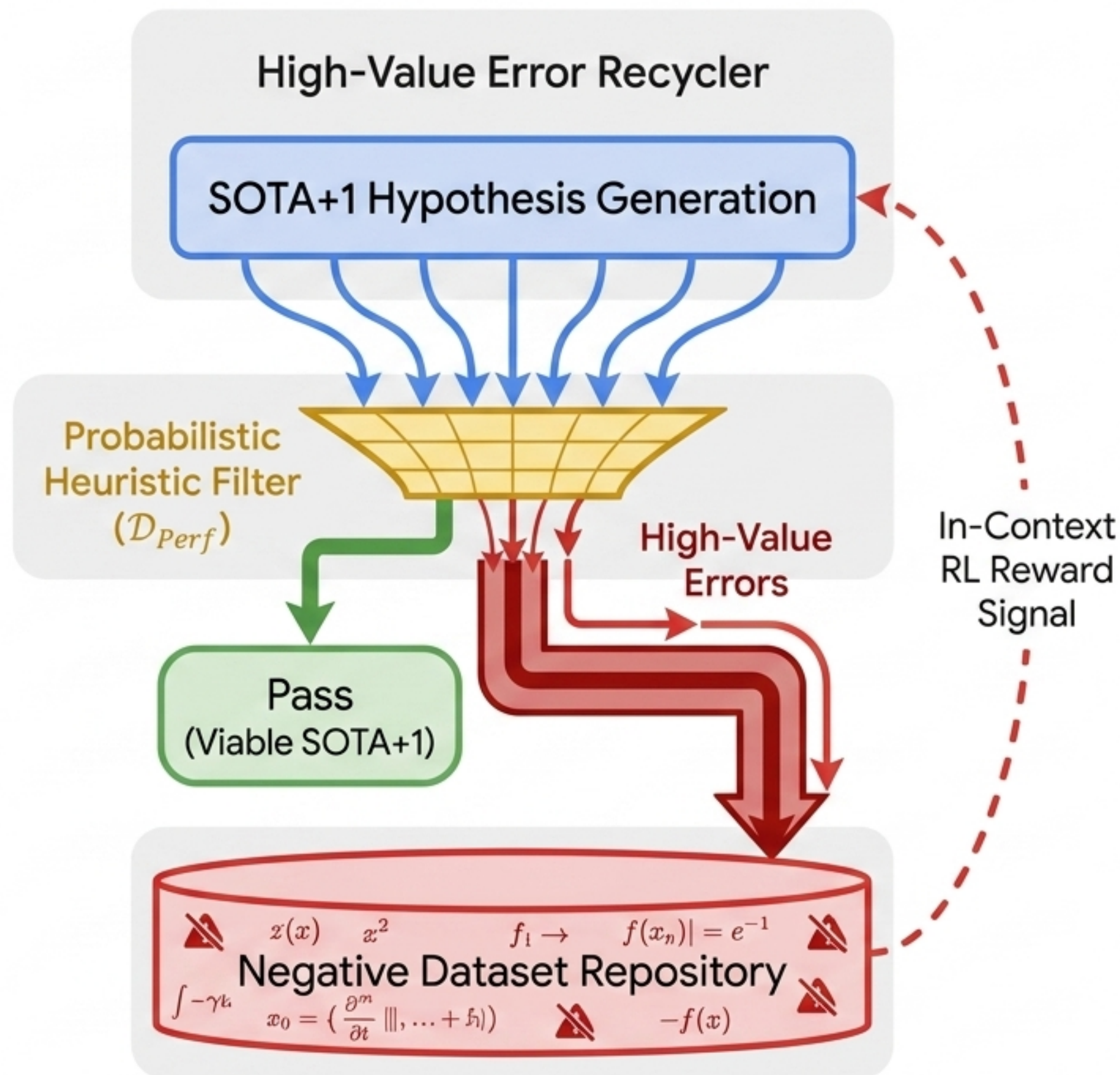
Phase 2: Empirical Execution

Gathers physical metrics to validate theoretical assumptions.

In-Silico Triage & Organic Reinforcement Learning

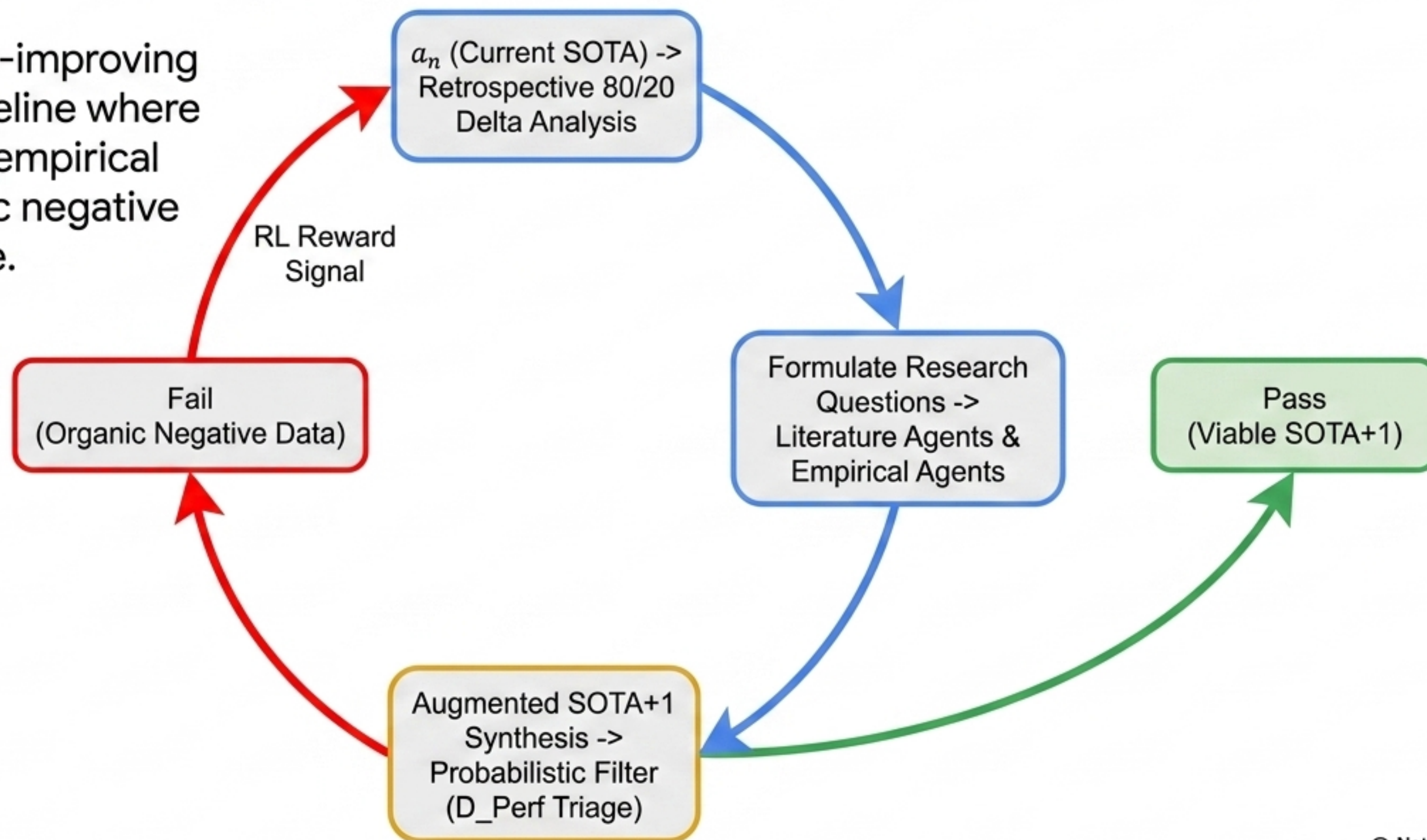
The Mechanism

Probabilistic generation naturally produces valuable failures. Instead of discarding them, ETFT uses these organic failures to continuously harden the **triage filter** and provide critical RL reward signals.

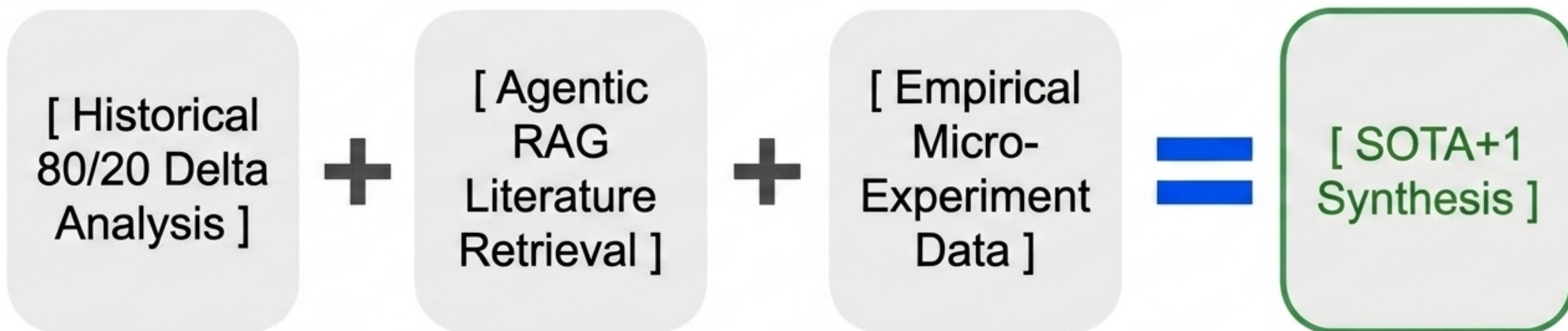


The Open-Loop Empirical ETFT Ecosystem

A continuously self-improving improving R&D pipeline where historical analysis, empirical inquiry, and organic negative feedback converge.



The SOTA+1 Equation



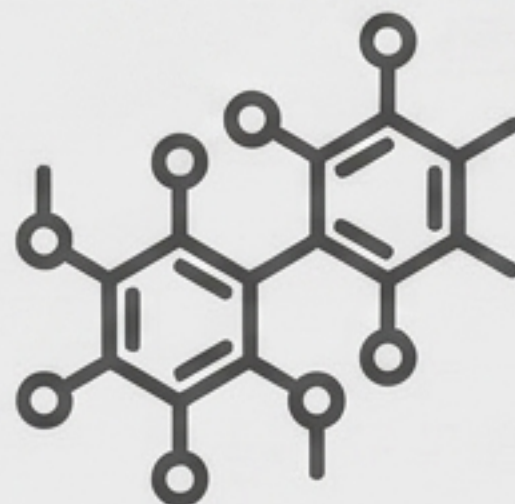
The General Innovation Accelerator

Core Concept: An algorithm is fundamentally any set of rules transforming inputs into outputs. ETFT is domain-agnostic—teaching machines not just what we know, but the very mechanics of how we invent.



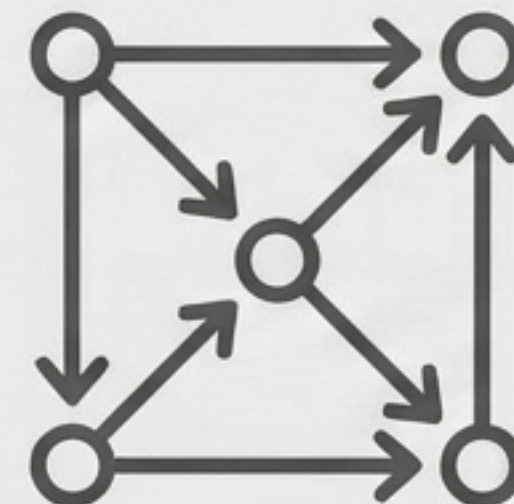
Materials Science

Evolution of Aerospace Alloys.



Pharmaceuticals

Progression of Targeted Therapies.



Operations

Optimization of Global Supply Chains.

Peer Review Validation & Next Steps



Verdict: Strong Accept (Recommended for Spotlight)

A masterclass in systems engineering... transcends traditional code-generation discourse to provide a scalable roadmap for an autonomous R&D scientist. – Official Peer Review Report, April 2026

Implementation Directives:

- Solve Multi-Agent Orchestration to prevent dead-locks in the ARL.
- Quantify the schema of D_Rationale to ensure accurate textual translation of code deltas.

Targeting: Proof of Concept for ICLR/NeurIPS.